

WORKED EXAMPLE

Momentum carries a signed displacement forward

The state is a parameter w plus a stored previous displacement v .

$$E(w) = \frac{1}{2}w^2, \quad g(w) = w, \quad w_0 = 2, \quad v_0 = 0$$

$$\eta = 0.1, \quad \beta = 0.9$$

Eta scales the current gradient. Beta retains 90 percent of the previous signed displacement, not 90 percent of an averaged gradient.

$$v_{k+1} = \beta v_k - \eta g(w_k), \quad w_{k+1} = w_k + v_{k+1}$$

A negative v moves the parameter downward when added. This convention matches the lecture signed Delta- W displacement. Do not add an extra $(1-\beta)$ factor to the gradient term.

WORKED EXAMPLE

Nesterov changes where the gradient is evaluated

The temporary look-ahead anticipates the movement from stored displacement.

$$q_k = w_k + \beta v_k$$

$$v_{k+1} = \beta v_k - \eta g(q_k), \quad w_{k+1} = w_k + v_{k+1}$$

q is a temporary gradient-evaluation point. It is not the stored parameter and is not an additional completed training update.

Lecture4 PDF page90 writes the signed displacement as Delta-W. The subscript k on v and w means the displacement and parameter after k updates.

$$w_{k+1} = q_k - \eta g(q_k)$$

This equivalent expression already includes beta times the old displacement through q . Do not add that contribution again.

WORKED EXAMPLE

Plain gradient descent gives a useful baseline

Reset to $w_0=2$ and take two updates without stored displacement.

$$g_0 = 2, \quad w_1 = 2 - 0.1(2) = 1.8$$

$$g_1 = 1.8, \quad w_2 = 1.8 - 0.1(1.8) = 1.62$$

$$E(w_2) = \frac{1}{2}(1.62)^2 = 1.3122$$

The next pages restart both momentum methods from the same original $w_0=2, v_0=0$. Their updates are alternatives to this baseline.

WORKED EXAMPLE

Momentum, update 1

Compute the gradient location first, then the new displacement, then the parameter.

$$w_0 = 2, \quad v_0 = 0$$

$$g_0 = 1(2) = 2$$

$$v_1 = 0.9(0) - 0.1(2) = -0.2$$

$$w_1 = 2 + (-0.2) = 1.8$$

The gradient uses current w . Keep the new displacement as state for the next update.

WORKED EXAMPLE

Momentum, update 2

Compute the gradient location first, then the new displacement, then the parameter.

$$w_1 = 1.8, \quad v_1 = -0.2$$

$$g_1 = 1(1.8) = 1.8$$

$$v_2 = 0.9(-0.2) - 0.1(1.8) = -0.36$$

$$w_2 = 1.8 + (-0.36) = 1.44$$

The gradient uses current w . Keep the new displacement as state for the next update.

WORKED EXAMPLE

Nesterov, update 1

Compute the gradient location first, then the new displacement, then the parameter.

$$w_0 = 2, \quad v_0 = 0$$

$$q_0 = 2 + 0.9(0) = 2$$

$$g_0 = 1(2) = 2$$

$$v_1 = 0.9(0) - 0.1(2) = -0.2$$

$$w_1 = 2 + (-0.2) = 1.8$$

The gradient uses the temporary q shown above; the final parameter uses current w plus the new displacement.

WORKED EXAMPLE

Nesterov, update 2

Compute the gradient location first, then the new displacement, then the parameter.

$$w_1 = 1.8, \quad v_1 = -0.2$$

$$q_1 = 1.8 + 0.9(-0.2) = 1.62$$

$$g_1 = 1(1.62) = 1.62$$

$$v_2 = 0.9(-0.2) - 0.1(1.62) = -0.342$$

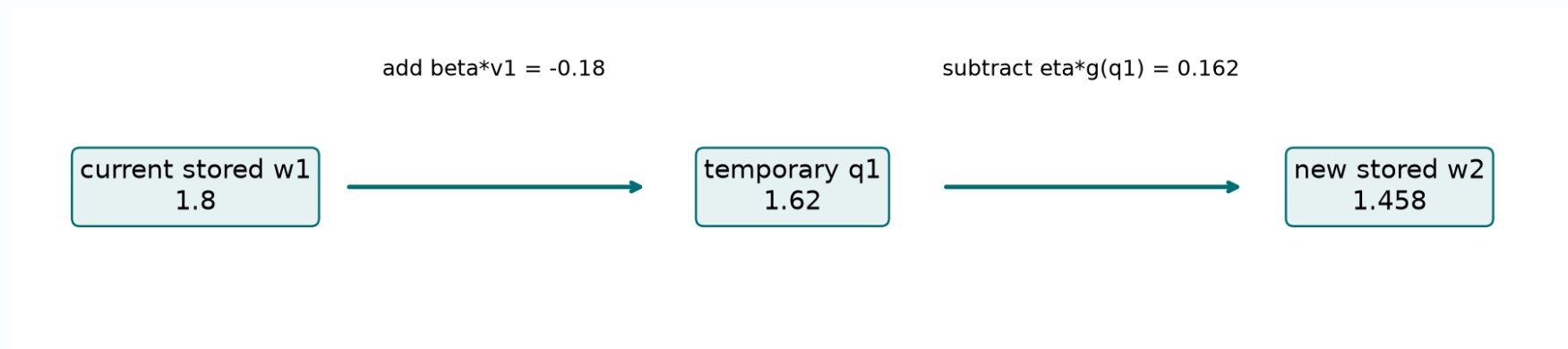
$$w_2 = 1.8 + (-0.342) = 1.458$$

The gradient uses the temporary q shown above; the final parameter uses current w plus the new displacement.

WORKED EXAMPLE

Trace Nesterov second-step dependencies

This picture is a calculation flow; q is temporary.



$$w_2 = q_1 - 0.1g(q_1) = 1.62 - 0.1(1.62) = 1.458$$

$$V_2 = w_2 - w_1 = 1.458 - 1.8 = -0.342$$

Using q_1 plus the entire new v_2 would count the old momentum contribution twice. Use either $w_1 + v_2$ or q_1 minus the fresh gradient correction.

WORKED EXAMPLE

Compare all second-step states

With zero initial displacement, both momentum methods share the first step.

method	gradient point	second gradient	new v	new w
GD	1.8	1.8	not stored	1.62
Momentum	1.8	1.8	-0.36	1.44
Nesterov	1.62	1.62	-0.342	1.458

$$E_{GD} = 1.3122, \quad E_M = \frac{1.44^2}{2} = 1.0368$$

$$E_N = \frac{1.458^2}{2} = 1.062882$$

This particular two-step example gives these losses. It does not establish that one method always wins. Changing curvature, rate, momentum coefficient or initial state can change the comparison.

INDEPENDENT PRACTICE

Your turn: new curvature and momentum coefficient

Reset all three methods before comparing them.

$$E(w) = w^2, \quad w_0 = 1, \quad v_0 = 0, \quad \eta = 0.1, \quad \beta = 0.5$$

Calculate two updates of plain gradient descent, momentum and Nesterov. Record every gradient-evaluation point and both momentum displacements.

Explain why the derivative is now $2w$ instead of w . In the Nesterov second step, distinguish temporary q from stored w .

Try the calculation before reading the next complete answer pages.

WORKED ANSWER

Practice: derivative and gradient-descent baseline

The missing factor 1/2 changes the derivative.

$$g(w) = \frac{d(w^2)}{dw} = 2w$$

$$g_0 = 2(1) = 2, \quad w_1 = 1 - 0.1(2) = 0.8$$

$$g_1 = 2(0.8) = 1.6, \quad w_2 = 0.8 - 0.1(1.6) = 0.64$$

$$E(w_2) = 0.64^2 = 0.4096$$

Both momentum methods also start at $w_0=1, v_0=0$. Their first update will match this first step because the old displacement is zero.

WORKED ANSWER

Practice: Momentum, update 1

Compute the gradient location first, then the new displacement, then the parameter.

$$w_0 = 1, \quad v_0 = 0$$

$$g_0 = 2(1) = 2$$

$$v_1 = 0.5(0) - 0.1(2) = -0.2$$

$$w_1 = 1 + (-0.2) = 0.8$$

The gradient uses current w . Keep the new displacement as state for the next update.

WORKED ANSWER

Practice: Momentum, update 2

Compute the gradient location first, then the new displacement, then the parameter.

$$w_1 = 0.8, \quad v_1 = -0.2$$

$$g_1 = 2(0.8) = 1.6$$

$$v_2 = 0.5(-0.2) - 0.1(1.6) = -0.26$$

$$w_2 = 0.8 + (-0.26) = 0.54$$

The gradient uses current w . Keep the new displacement as state for the next update.

WORKED ANSWER

Practice: Nesterov, update 1

Compute the gradient location first, then the new displacement, then the parameter.

$$w_0 = 1, \quad v_0 = 0$$

$$q_0 = 1 + 0.5(0) = 1$$

$$g_0 = 2(1) = 2$$

$$v_1 = 0.5(0) - 0.1(2) = -0.2$$

$$w_1 = 1 + (-0.2) = 0.8$$

The gradient uses the temporary q shown above; the final parameter uses current w plus the new displacement.

WORKED ANSWER

Practice: Nesterov, update 2

Compute the gradient location first, then the new displacement, then the parameter.

$$w_1 = 0.8, \quad v_1 = -0.2$$

$$q_1 = 0.8 + 0.5(-0.2) = 0.7$$

$$g_1 = 2(0.7) = 1.4$$

$$v_2 = 0.5(-0.2) - 0.1(1.4) = -0.24$$

$$w_2 = 0.8 + (-0.24) = 0.56$$

The gradient uses the temporary q shown above; the final parameter uses current w plus the new displacement.

WORKED ANSWER

Practice checkpoint and common mistakes

The state you carry changes the next update.

method	second gradient point	second gradient	final w
GD	0.8	1.6	0.64
Momentum	0.8	1.6	0.54
Nesterov	0.7	1.4	0.56

$$E_M = 0.54^2 = 0.2916, \quad E_N = 0.56^2 = 0.3136$$

Do not reset velocity after every step, use current-w gradient for Nesterov, add the old momentum twice, or mix this displacement convention with a gradient-average formula containing (1-beta).

Changing the definition of stored state can give an equivalent algorithm only with the corresponding sign and learning-rate conversion. Match the stated convention before substituting numbers.